

Build a Human Classifier

Project: Helpful Classroom Sorter

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Learning from Examples

Without opening the old lesson, explain one key idea from Learning from Examples. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Repeating Efficiently

Without opening the old lesson, explain one key idea from Repeating Efficiently. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Features: Useful Clues

Without opening the old lesson, explain one key idea from Features: Useful Clues. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Construct both outcome paths for Constructing a classifier. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Constructing a classifier. Proposed explanation: “Any numeric input must receive a category”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Build an independent ribbon classifier

Design Ribbon1 for ribbon lengths recorded as whole centimetres from 1 through 12 inclusive, with unit cm required. Predict long at 8 cm or more and short otherwise. Write complete instructions and trace 7 cm, 8 cm, a blank length and 13 cm. Include reasons for no prediction and the model version in each record.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. Your learning target

A classroom rule labels anything with wings as a bird. Test a bat, a penguin and a toy aeroplane. Define the target independently, then propose a next step without pretending one visual clue solves biology.

6. Solve the given case

A human classifier uses has wheels means vehicle on a bicycle, wheelchair, toy car and boat.

Apply the rule and identify a limitation.

Show your trace, calculation or evidence

Write or draw your reasoning and evidence.

7. Check a changed case

Use a human classifier: three sides means triangle; four equal sides means square. Classify a three-sided card and a card whose corner is hidden. When should you pause?

My independent answer and reason

Write or draw your reasoning and evidence.

Week 14 Test and reflect

8. Design a revealing test

Create a changed input, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record · My evidence

Test record: Changed input or condition

My evidence: _____

Test record: Expected answer or fallback

My evidence: _____

Test record: Observed result

My evidence: _____

Test record: Why the result occurred

My evidence: _____

9. Project checkpoint

Make the sorter's current decision boundary visible as a rule chart or feature map.

The project change and evidence I will keep

Write or draw your reasoning and evidence.

10. Self check

I can explain the mechanism. I can show evidence. I can handle a changed case. I can name a limit. Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

11. Transfer practice from the original programme

A classroom rule labels anything with wings as a bird. Test a bat, a penguin and a toy aeroplane. Define the target independently, then propose a next step without pretending one visual clue solves biology.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

12. W1 Write an input contract

Write the complete Tower1 input contract. Give one accepted example and three different rejected examples, each with its reason. Explain why an accepted input does not have to be one of the examples used to fit the threshold.

My reasoning and evidence

13. W2 Construct both outcome paths

Write numbered instructions for Tower1, including input checks, the stopping path after a failed check, the threshold rule and the record to keep. Your partner must be able to use them without supplying an unstated rule.

My reasoning and evidence

14. W3 Trace boundaries and a broken path

Trace 1 cm, 5 cm, 6 cm, 10 cm and 11 cm using Tower1. A faulty procedure records no prediction for a blank, then continues and treats the blank as zero to report short. Identify the missing instruction and the proper final outcome.

My reasoning and evidence

15. W4 Build an independent ribbon classifier

Design Ribbon1 for ribbon lengths recorded as whole centimetres from 1 through 12 inclusive, with unit cm required. Predict long at 8 cm or more and short otherwise. Write complete instructions and trace 7 cm, 8 cm, a blank length and 13 cm. Include reasons for no prediction and the model version in each record.

My reasoning and evidence

16. W5 Explain what your procedure establishes

Ribbon1 accepts 12 cm and predicts long. No comparison label has been supplied for that record. State what has been checked and what remains unknown. Explain why changing a blank to 1 cm just to produce a category would be a mistake.

My reasoning and evidence

17. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

18. Pictorial case practice: Build a Human Classifier

A fictional ribbon sorter uses a fixed instruction card called RibbonR.

Valid input: whole-number length from 0 to 15 with unit cm.

Predict long at least 9 cm; otherwise short. Missing, wrong-unit or out-of-range values request clarification.

Build a Human Classifier

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Input contract

Check that length is present, whole, within 0-15, and in cm.



2. Valid cases

8 cm -> short; 9 cm -> long.



3. Invalid cases

16 cm and blank length -> clarification, not a forced label.



4. Keep records

Store input, contract result, rule version and prediction.

CHANGE ONE THING

Provide length 9 with unit mm.

Explain the changed result using the matching lesson.

Trace 0 cm and explain why clarification is different from predicting short.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: Create a sorter instruction card covering valid values, equality, missing fields and who can resolve clarification requests.

19. Project portfolio milestone

Project: Helpful Classroom Sorter

Current milestone: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

Create a sorter instruction card covering valid values, equality, missing fields and who can resolve clarification requests.

Evidence to keep · My record

Evidence to keep: This week’s input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week’s record rather than starting again.

20. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional ribbon sorter uses a fixed instruction card called RibbonR.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Store input, contract result, rule version and prediction.

1. Invalid cases
2. Input contract
3. Keep records
4. Valid cases

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Keep records Store input, contract result, rule version and prediction. A prediction boundary answers a different question from input validation. Length 9 passes the contract and meets the long boundary. Length 16 must not receive long merely because it exceeds 9: it violates the allowed range first. Zero is permitted and receives short.

Practice 2: explain the deciding evidence

Changed-case practice: Provide length 9 with unit mm. Which conclusion is supported by the supplied facts?

1. Request clarification rather than predicting long.
2. This one example guarantees correct results for every future input.
3. We can decide the result without examining the changed input or the procedure.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: Request clarification rather than predicting long. RibbonR requires cm; the unit mismatch fails its declared input contract.

Repair a missing building block

Review Learning from Examples (ai-1/13). Goal: Fit a threshold by tracing predictions and counting disagreements for every candidate. Short practice: Which new measured length would reveal disagreement between the candidates?. Return to this week and explain what you changed.

Review Repeating Efficiently (ai-1/11). Goal: Trace a loop by testing its condition before every repetition. Short practice: Trace a fresh start of 9. How many additions occur, and must the report equal 10?. Return to this week and explain what you changed.

Review Features: Useful Clues (ai-1/8). Goal: Choose recorded properties that preserve distinctions needed for a stated task. Short practice: Point to the pair that tests pattern, then explain why the sticker is a bad early input.. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
